Practical No.: 3

Name: Nikita Lalaso Narute **Roll No.: COB422**

Batch: B2 **Div: B**

Class: BE COMPUTER

Title:

Write a program to solve a fractional Knapsack problem using a greedy method.

Program:

```
#include <bits/stdc++.h>
using namespace std;

struct Item {
    int profit, weight;
    Item(int profit, int weight) {
        this->profit = profit;
        this->weight = weight;
    }
};

static bool cmp(Item a, Item b) {
    double r1 = (double)a.profit / a.weight;
    double r2 = (double)b.profit / b.weight;
    return r1 > r2;
}

double fractionalKnapsack(int W, Item arr[], int N) {
    sort(arr, arr + N, cmp);
    double finalvalue = 0.0;

    for (int i = 0; i < N; i++) {
        if (arr[i].weight <= W) {
            W -= arr[i].weight;
            finalvalue += arr[i].profit;
        } else {
            finalvalue += arr[i].profit *
                          ((double)W / arr[i].weight);
            break;
        }
    }
    return finalvalue;
}

int main() {
    int W = 50;
    Item arr[] = {{60, 10}, {100, 20}, {120, 30}};
    int N = sizeof(arr) / sizeof(arr[0]);

    cout << fractionalKnapsack(W, arr, N) << endl;
    return 0;
}
```

Output:

```
(base) sspm@sspm:~$ cd Desktop
(base) sspm@sspm:~/Desktop$ g++ daa3.cpp
(base) sspm@sspm:~/Desktop$ ./a.out
240
(base) sspm@sspm:~/Desktop$
```